FOR THE TUTOR OR PARENT

Answer key: Ages 10-12

Accept equivalent forms (for example 70 cents and 70c, or 3/4 and three quarters). One mark per question. Add the warm-up and main marks to get a score out of 10; the Brain stretchers are a bonus and are not counted. Keep this key away from the child until they have finished.

Warm-up (4 marks)

#	Question	Answer	Watch for
1	What is 5/6 - 1/3? Give your answer as a fraction in simplest form.	**1/2**	Look for converting 1/3 to 2/6, then simplifying 3/6. Common errors: subtracting numerators and denominators (4/3) or leaving 3/6.
2	A ribbon 2.4 m long is cut into 6 equal pieces. How long is each piece, in metres?	**0.4**	Check place value; common error is 0.04 or 4 from misplacing the decimal point.
3	What is 3/4 of 48?	**36**	Look for 48 / 4 = 12, then 12 x 3. Common errors: 12 (divided by 4 only) or 192 (multiplied).
4	A rectangle has a perimeter of 36 cm and a length of 11 cm. What is its area in square centimetres?	**77**	Child should find width 7 cm first (36 / 2 - 11). Common error: using 36 - 22 = 14 as the width or answering 7.

Main questions (6 marks)

#	Question	Answer	Watch for
5	Round 4,386,512 to the nearest hundred thousand.	**4400000**	Look at the ten-thousands digit (8) and round up. Accept 4 400 000, 4,400,000 or 4.4 million. Common errors: 4,300,000 (truncating) or 4,390,000 (wrong place).
6	Sam spent 2/5 of his money on a book. He then spent 1/3 of the remainder on lunch. He had $24 left. How much money did he have at first, in dollars? (Not scored: how did you start? Write one line or draw a quick picture.)	**60**	Best seen with a bar model: remainder is 3/5, lunch is 1/5 of total, left 2/5 = $24. Common error: treating 1/3 as a fraction of the whole.
7	The area of a triangle is 40 square cm. Its base is 16 cm. What is its height in cm?	**5**	Child should use area = 1/2 x base x height. Common error: forgetting the 1/2 and answering 2.5, or dividing 40 by 16 only.
8	A tank is 50 cm long, 30 cm wide and 40 cm high. A tap fills it at 6 litres per minute. How many minutes to fill it completely? (1 litre = 1000 cubic cm) (Not scored: how did you start? Write one line or draw a quick picture.)	**10**	Volume 60,000 cubic cm = 60 L. Common errors: not converting to litres, or adding the dimensions.
9	In a class of 40 pupils, 60% are girls. 25% of the girls wear glasses. How many girls wear glasses?	**6**	Girls = 24, then a quarter = 6. Common error: taking 25% of 40 (10).
10	In triangle ABC, AB = AC and angle BAC = 50 degrees. What is angle ABC, in degrees?	**65**	Look for (180 - 50) / 2. Common errors: 130 (forgot to share the rest between two equal angles) or 50.

Brain stretchers (bonus, not counted)

#	Question	Answer	Watch for
11	Look at the pattern 2, 5, 10, 17, 26, ... What is the 10th number in the pattern?	**101**	Look for differences 3,5,7,9 or the rule n x n + 1. Common error: guessing a constant difference.
12	Ali has 3 times as many stickers as Ben. After Ali gives 18 stickers to Ben, Ali has twice as many as Ben. How many stickers did Ali have at first?	**162**	Bar-model or before-after reasoning. Ben starts with 54. Common error: answering Ben's amount (54) or ignoring the change in total.

Tutor note. Finished very fast? Ask the child to explain one answer aloud before you mark: a fast wrong answer is often a skipped step, not a missing topic. The parent makes the final choice.

Questions the child ticked "not learnt yet" score 0 but tell you what to teach first. They do not count against the child.